A-A-53845
10 April 1989

COMMERCIAL ITEM DESCRIPTION

BURS, DENTAL, FINISHING (Tungsten Carbide)

The General Services Administration has authorized the use of this Commercial Item Description in preference to Military Specification MIL-B-37036A dated 30 December 1985.

This commercial item description covers tungsten carbide finishing burs suitable for use with friction grip handpieces in finishing dental restorations.

Salient characteristics:

Head shapes and size numbers. The tungsten carbide finishing burs shall be of the following head shapes and size numbers as specified below (see notes).

Round
Size No. 7002
Size No. 7004
Size No. 7008
Size No. 9004

Flame
Size No. 7106

Cone
Size No. 7204
Size No. 7206

Inverted Taper
Size No. 7304

Egg
Size No. 7406

Tapered
Size No. 7702
Size No. 9713

Bullet
Size No. 7801
Size No. 7803
Size No. 9803

Needle
Size No. 7901
Size No. 7902

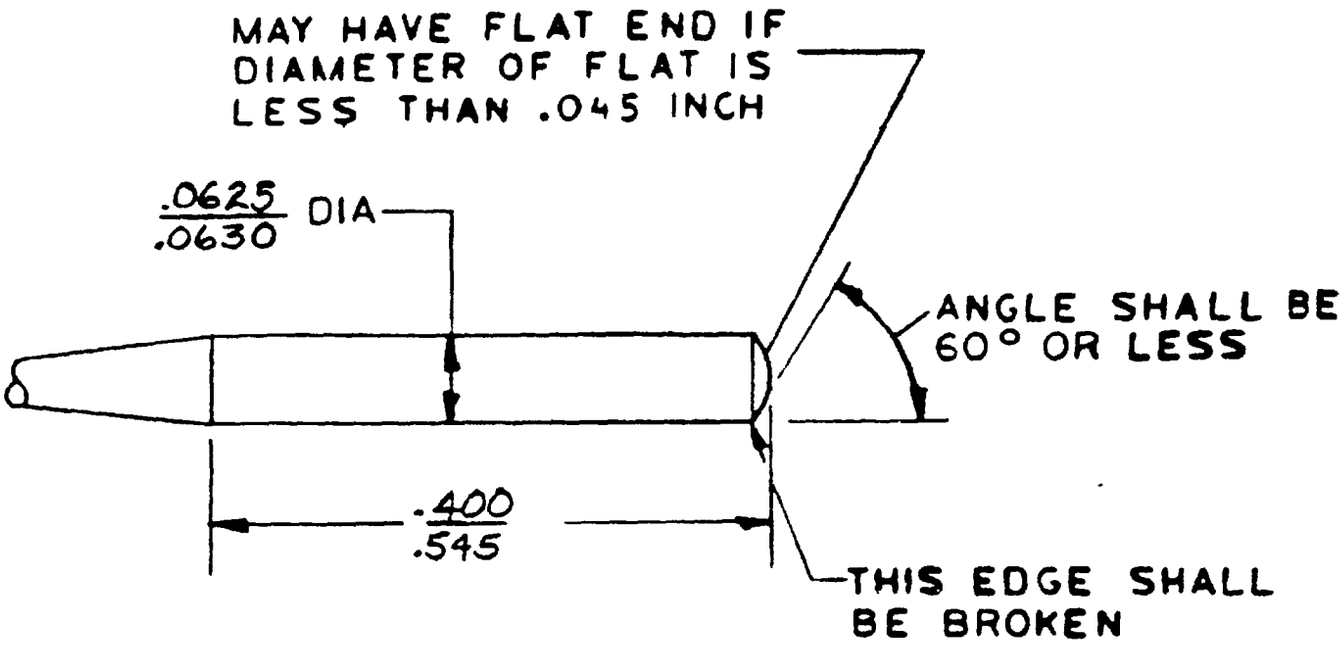
Material.

Head. The bur heads shall be fabricated from a solid piece of cemented tungsten carbide with cobalt binder. Titanium or tantalum carbide, or both, may be added to achieve the required properties.

AMSC N/A FSC 6520

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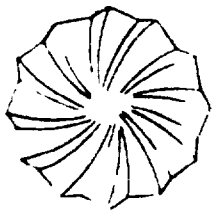


DIMENSIONS ARE IN INCHES.

FIGURE 1.

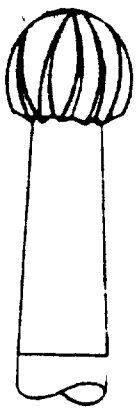
SHAPE AND DIMENSIONS OF BUR SHANK

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12 BLADES

CUTTING END
(TYPICAL)



ROUND



FLAME



CONE



INVERTED TAPER



EGG



TAPERED



BULLET



NEEDLE

FIGURE 2 - SHAPE OF BUR HEADS

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Shank, neck and base of bur. The shank, neck and the base of the burs shall be fabricated from S5 tool steel as specified in ASTM A681 for composition only or class 4 type 420C corrosion-resisting steel as specified in ASTM F899 for composition only.

Bur shape and dimensions.

Bur shank. The shape and dimensions of the bur shank shall be in accordance with Figure 1.

Bur head, neck and overall length of bur. The bur head shall be in accordance with Figure 2 for the indicated shape. The bur head, neck and overall length of the burs shall be in accordance with the dimensional requirements shown in Table 1 for the shape specified.

TABLE I. Head, neck and overall length dimensions.

Shape	Size No.	Head diameter at largest place ± .003 inch	Minimum length of head, inch	Maximum diameter of neck at head-end inch	Overall length of bur, inch
Round	7002	0.039	0.026	0.030	0.755 ± 0.035
Round	7004	0.055	0.032	0.036	0.755 ± 0.035
Round	7008	0.091	0.063	0.052	0.755 ± 0.035
Round	9004	0.055	0.032	0.035	0.748 ± 0.020
Flame	7106	0.071	0.135	0.057	0.755 ± 0.035
Cone	7204	0.055	0.335	see note	0.787 ± 0.035
Cone	7206	0.071	0.335	see note	0.787 ± 0.035
Inverted Taper	7304	0.055	0.161	0.040	0.787 ± 0.035
Egg	7406	0.071	0.130	0.057	0.755 ± 0.035
Tapered	7702	0.040	0.193	0.040	0.787 ± 0.035
Tapered	9713	0.047	0.200	0.040	0.866 ± 0.020
Bullet	7801	0.035	0.100	0.035	0.755 ± 0.035
Bullet	7803	0.048	0.110	0.048	0.755 ± 0.035
Bullet	9803	0.047	0.110	0.039	0.748 ± 0.020
Needle	7901	0.036	0.126	0.035	0.755 ± 0.035
Needle	7902	0.040	0.130	0.040	0.755 ± 0.035

NOTE: Cone does not have a neck.

The number of blades shall be 12 for all burs, except sizes 9004, 9713 and 9803 shall be 30 blades. For burs 7204 and 7206, the taper or cone angle shall be 8-16, and for burs 7702 and 9713, it shall be 4-11 (the total for both sides of the angle).

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Design and construction. The burs shall be dental finishing burs, suitable for use in friction grip handpieces in trimming and finishing dental restorations. Burs shall consist of a shank, a neck (except for the cone which does not have a neck), and a head. The blades shall be uniformly cut and sharpened. The burs shall be of two-piece construction. The joint of burs shall be permanently welded or brazed to the shank. The welded joint shall be milled so that the resultant shape has the appearance of an integral head unit.

Runout. The total indicated runout of the bur shall not exceed 0.002 inch, when tested as specified herein.

Strength of neck and joint. The neck shall not break on any of the burs, and the burs shall not fracture, when tested as specified herein. The nondamaging test load shall be as specified in Table II for the applicable bur size number.

TABLE II. Strength requirements.

Bur size number	Nondamaging test load, pounds
7002	3.5
7004	6.5
7008	8.5
7106	6.5
7204	4.0
7206	6.0
7304	2.5
7406	6.5
7702	2.0
9713	4.5
7801	3.0
7803	4.0
9803	5.5
7901	2.0
7902	3.0
9004	6.5

Increase in runout. The increase in runout shall be no greater than 0.002 inch, when determined as specified herein.

Autoclaving. The bur shall show no evidence of corrosion, after being subjected to the autoclave test as specified herein, except that corrosion present at the welded joint shall be disregarded, and shall not be cause for rejection.

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Surface roughness. The surface roughness of the shanks of the burs shall be no more than the root-mean-square (RMS) average deviation of 20 microinches in the longitudinal direction of the shank, when determined as specified herein.

Finish. The bur head shall have a finely honed finish with sharpened blades. The neck and shank of the burs shall be smooth. The tool steel portion of the burs shall be nickel or chromium plated. In addition, a flash of gold plate on the shank shall be optional.

Identification marking. Marking of the bur with the name or registered trademark shall be at the option of the manufacturer.

Workmanship. The burs shall be free from defects which may detract from their appearance or impair their serviceability.

Unit. Package. (PG) One package containing ten burs of one shape and size only, as specified, constitutes one unit. Each shall be packaged in a plastic holder, so designed that each bur is contained in an individual cell.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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Certificates of quality. Certificates of quality, supplied by the manufacturer of the steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same bur shall be used for the determination of two or more inspection and test characteristics. The unit of product for inspection, examination and test purposes shall be one package of (10 burs). Unless otherwise specified, sampling for examination shall be conducted at inspection level S-2, except for dimensions which shall be conducted at inspection level S-1, AQL 2.5 (percent defective), and sampling for tests (when applicable) shall be conducted at inspection level S-1. For tests, acceptance number shall be zero (0) for all sample sizes.

Raw material. A sufficient quantity of each lot of raw material used in the manufacture of the burs shall be selected for chemical analysis.

Examination. Examination shall include but shall not be limited to the following defects:

Defects.

Bur size number or head shape not as specified.
Head diameter exceeds specification requirements at maximum cross-section.
Length of head less than minimum requirement for applicable bur size number.
Cutoff end on shank not as shown in figure 1.
Burs not free from defects such as chipped blades, cracks, or pits.
Number of blades not as specified.
Finish on bur not as specified.
Overall length of bur not within specification limits.
Maximum neck diameter at head end exceeds specification limits.
Shank dimensions not within specification limits.
Taper or cone angle of head (when applicable) not within specification limits.
Welded joint of two-piece bur does not give the appearance of an integral head unit.
Packaging and packing not as specified.
Marking on packaging or pack incorrect, illegible, incomplete, or missing altogether.

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Dimensions. Examination of the dimensions shall be made with a toolmaker's microscope, or other instrument of equal accuracy. Manufacturer's gage may be used if properly calibrated checking standards are used in conjunction with them.

Microscope examination. Examination of bur defects not readily perceptible to the naked eye shall be made using a microscope at 10 power (10x) magnification.

Tests.

Chemical composition. In the absence of certificates of quality for the raw material, the samples selected shall be analyzed in accordance with Method 111 or 112 of FED-STD-151. In case of dispute, Method 111 shall be used for referee analysis. (Separate analysis shall be made on the carbide and the non-carbide portions of two-piece burs).

Runout. The bur shall be placed and held in a "V" block or precision chuck, and rotated slowly. A dial indicator sensitive to 0.0002 inch and equipped with a knife-edge foot, or a shadowgraph or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the bur, and so positioned that the foot contacts the bur neck at the point of smallest diameter. For shape C burs, the foot shall contact the shank at the point nearest the head. The maximum and minimum readings obtained during one complete 360-degree rotation of the bur shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

Neck and joint strength.

Equipment for test. The equipment shall consist of a true-running chuck suitable for holding and rotating the test bur, and a loading device capable of applying the load indicated in Table II through an anvil with a flat surface at an angle of 22-1/2 $\pm$ 1/2 degrees to the bur axis. The chuck shall grip at least a 3/8 inch portion of the bur shank starting at the intersection of the neck and shank.

Test procedure. Following an initial examination for total indicated runout as specified above, those burs which have been selected for the neck and joint strength test shall be rotated in the chuck to that position which shows the highest deflection reading on the dial indicator gage.

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A load of the magnitude specified in Table II shall be slowly applied at an angle to the bur axis of 22-1/2 ± 1/2 degrees, in the direction which tends to further increase the dial deflection reading. The neck and joint of the bur shall be observed for evidence of breakage. Should breakage not have occured at the point of maximum deflection, the load shall be removed and the examination for increase in runout determined as specified below.

Increase in runout. The increase in runout shall be determined only on those samples which have been subjected to the neck and joint strength, and shall be performed in the manner already specified for runout. The difference between the final and the initial indicated runout shall be taken as the increase in runout.

Autoclaving. The test burs shall be cleaned in accordance with the instructions supplied by the manufacturer. The burs shall then be autoclaved at 270 ± 5 degrees Fahrenheit (F.) and 27 ± 1 pounds per square inch pressure for a minimum of 20 minutes. After the burs are removed from the autoclave, the entire surface area of the non-carbide portion of the bur shall be examined for evidence of corrosion.

Surface roughness. The surface roughness of the bur shall be determined by using a Brush Surfindicator, or other equipment of equal accuracy. The RMS average shall be determined by squaring the height measurements of the peaks and valleys of the surface of the shank, obtaining the averages of the squared values, and then extracting the square root of this average square.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerance specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquistion Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

NSN	Item Identification	Specification Shape	Designation Size No.
6520-01-098-5807	Bur, Dental, Finishing, Friction Grip Angle Handpiece, Tungsten Carbide, No. 7002, 10s	Round	7002
6520-01-098-5808	Bur, Dental, Finishing, Friction Grip Angle Handpiece, Tungsten Carbide, No. 7304, 10s	Inverted Taper	7304

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NSN	Item Identification	Specification Shape	Designation Size No.
6520-01-098-5809	Bur, Dental, Finishing, Friction Grip Angle Handpiece, Tungsten Carbide, No. 7406, 10s	Egg	7406
6520-01-098-5810	Bur, Dental, Finishing, Friction Grip Angle Handpiece, Tungsten Carbide, No. 7702, 10s	Tapered	7702
6520-01-098-5811	Bur, Dental, Finishing, Friction Grip Angle Handpiece, Tungsten Carbide, No. 7803, 10s	Bullet	7803
6520-01-098-5812	Bur, Dental, Finishing, Friction Grip Angle Handpiece, Tungsten Carbide, No. 7902, 10s	Needle	7902
6520-01-098-5813	Bur, Dental, Finishing, Friction Grip Angle Handpiece, Tungsten Carbide, No. 7106, 10s	Flame	7106
6520-01-098-5814	Bur, Dental, Finishing, Friction Grip Angle Handpiece, Tungsten Carbide, No. 7008, 10s	Round	7008
6520-01-098-5815	Bur, Dental, Finishing, Friction Grip Angle Handpiece, Tungsten Carbide, No. 7004, 10s	Round	7004
6520-01-098-5816	Bur, Dental, Finishing, Friction Grip Angle Handpiece, Tungsten Carbide, No. 7204, 10s	Cone	7204

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NSN	Item Identification	Specification Shape	Designation Size No.
6520-01-098-5817	Bur, Dental, Finishing, Friction Grip Angle Handpiece, Tungsten Carbide, No. 7206, 10s	Cone	7206
6520-01-099-0328	Bur, Dental, Finishing, Friction Grip Angle Handpiece, Tungsten Carbide, No. 7901, 10s	Needle	7901
6520-01-105-0429	Bur, Dental, Finishing, Friction Grip Angle Handpiece, Tungsten Carbide, No. 7801, 10s	Bullet	7801
6520-01-105-0430	Bur, Dental, Finishing, Friction Grip Angle Handpiece, Tungsten Carbide, No. 9004, 10s	Round	9004
6520-01-273-2587	Bur, Dental, Finishing, Friction Grip Angle Handpiece, Tungsten Carbide, No. 9803, 10s	Bullet	9803
6520-01-273-2588	Bur, Dental, Finishing, Friction Grip Angle Handpiece, Tungsten Carbide, No. 9713, 10s	Tapered	9713

Ordering data. Intermediate/exterior package quantities and labeling and marking must be as specified in the contract and/or order.

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MILITARY INTERESTS:

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Navy - MS
Air Force - 03

CIVIL AGENCY COORDINATING ACTIVITIES:
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05971

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

PROJECT NO. 6520-2221

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